

Technical Datasheet WIN-SN-TnH-IAQ-W Sensor

The Sensor Unit collects the environmental data of 3 parameters from the indoor surrounding and displays the data as well as sends the data to the Cloud over MQTT protocol over WI-FI. The unit is robustly designed for continuous 24/7 operation and is exceptionally easy to use.

Display Module: -



The module is housed in a rugged industrial enclosure with wall mounting slots featuring an integrated display for easy installation deployment. Configuration settings such as Wi-Fi connectivity, MQTT parameters and cloud communication can be managed through the integrated display interface or the built-in web portal.

The modules are designed for reliable long-term operation with built-in protection, ensuring enhanced safety and system reliability. The Display Unit collects, displays, and securely publishes sensor data to the cloud using the MQTT protocol, enabling continuous remote monitoring and data access.

Specifications

General –

Dimensions	98 mm L x 98 mm B x 34 mm H
Power	Input Power – 5 VDC Type C
Operating Temperature	-20°C to +80°C (-4°F to +176°F)
Storage temperature	-20°C to +80°C (-4°F to +176°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



Operator Interface & Navigation Guide

1. Initial Network Provisioning (Wi-Fi Setup):

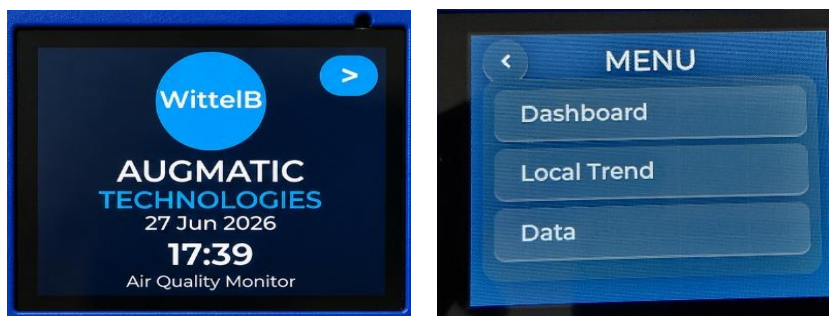
Upon first boot—or when a network change is required—the device enters Access Point (AP) mode. The screen displays a "Wi-Fi Setup Required" prompt, broadcasting its own local network. Users simply connect to this network via a mobile device or laptop using the **Password 12345678** and navigate to the provided IP address (ex: 192.168.4.1) to securely pass local Wi-Fi credentials to the unit.



2. Home Screen & Main Navigation

Once provisioned, the device boots to the Home Screen, displaying the company logo, current date/time, and system title. Tapping the [>] icon in the top right corner opens the **Main Menu**, which splits the interface into two primary pathways:

- **Dashboard:** For live data of the sensors.
- **Local Trends:** Displays historical trend graphs for each monitored parameter (**Temperature, Humidity, and IAQ**). Users can independently select **6-hour, 12-hour, or 24-hour** trend data for each parameter.
- **Data:** Configure the device whenever the Wi-Fi network changes.



3. Live Telemetry Dashboard

The Dashboard provides a clear, real-time view of all monitored parameters on a single screen, enabling users to quickly assess the current environmental conditions.

- **Real-Time Monitoring:** Displays the latest values of all connected sensors, including **Temperature, Humidity, IAQ**
- **Color-Coded Display:** Each parameter is presented in a dedicated color-coded tile for improved readability and quick identification.
- **Instant Data Updates:** Sensor readings are refreshed continuously, providing up-to-date information for effective environmental monitoring.
- **User-Friendly Interface:** The intuitive display allows users to easily view and monitor all air quality parameters from a single dashboard screen.



4. Local Trends

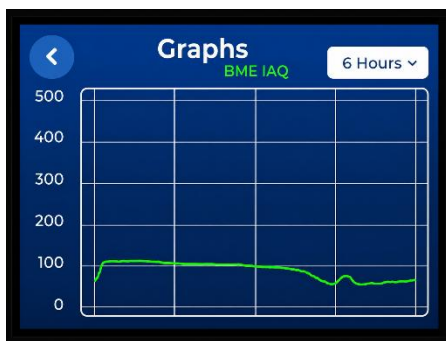


Image: 1

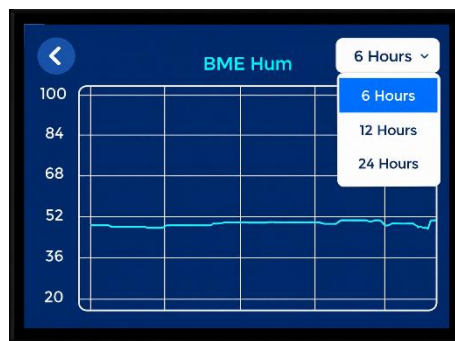


Image: 2

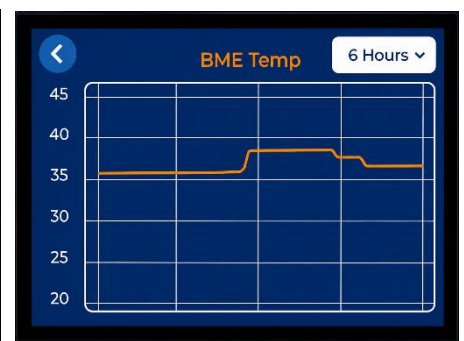


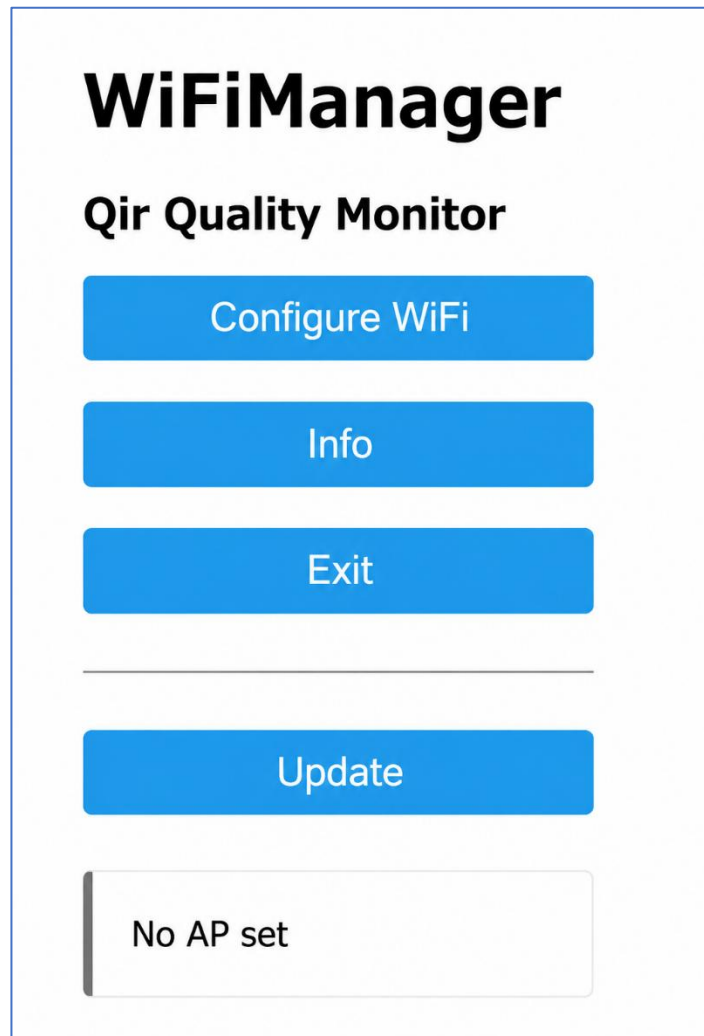
Image: 3

- **Live Trend Monitoring:** View real-time graphical data for **Temperature, Humidity, and IAQ**, enabling continuous monitoring of environmental conditions.
- **Individual Parameter Analysis:** Each parameter (**Temperature, Humidity, and IAQ**) can be viewed independently with its own dedicated trend graph.
- **Selectable Time Range:** Users can choose **6-hour, 12-hour, or 24-hour** historical data for each parameter individually to analyze short-term and long-term trends.
- **Default Time Range:** By default, the trend graph displays the **last 6 hours** of data. Users can change the view to **12 hours** or **24 hours** for each parameter individually.
- **CSV Data Export:** Download historical data in **CSV format** for the selected parameter and selected time range (**6 Hours, 12 Hours, or 24 Hours**) for offline analysis, reporting, and record keeping.

Web Portal Configuration Guide

1. Accessing the Wi-Fi Manager

- Navigating to the device's IP address opens the main WIFI Manager portal.
- Click the **[Configure Wi-Fi]** button to proceed to the network selection page.



2. Selecting the Network

- The setup page displays a list of available Wi-Fi networks in range.
- Tap the desired network from the list to automatically populate the **SSID input field** or type the SSID manually.
- Enter the network security key into the **Password input field**.
- Click the **Save button** to commit the credentials.
- The device will then process the information and connect to the assigned Wi-Fi network.

WittelB-SN-ATH-SN5	 
WittelB-SN-ATH-SN2	 
AirFiber-DIqCcP	 
DIRECT-FE-HP Smart Tank 750	 
WittelB	 
WittelB-SN-ATH-SN1	 
WIN-SLD	 
WittelB-SN-TH-SN7	 

SSID

Password

☐ Show Password

Save

Refresh

No AP set

Web Portal Configuration Guide (Air Quality Monitor)

The embedded web portal is dedicated to configuring Wi-Fi and MQTT cloud settings, enabling seamless cloud connectivity and remote sensor data monitoring.

1. Accessing the Settings

- Enter the IP address **192.168.1.60**.
- Then, you will be redirected to the cloud Settings page, where you can enter the cloud configuration details to publish the data to the MQTT Cloud.

AIR QUALITY MONITOR

Configuration Settings

WiFi Settings

Connected WiFi

IP Address

Device ID

MQTT Settings

MQTT Broker

Port

Security

None

Topic

Username

Password

Publish Interval

Seconds

SAVE CONFIGURATION

RESET SETTINGS

Instructions

1. Verify Connected WiFi and IP Address.
2. Enter MQTT Broker details and Port Number.
3. Select Security Mode (None/TLS).
4. Configure Topic, Device ID and Publish Interval.
5. Press SAVE CONFIGURATION.
6. Device will restart and connect automatically.
7. Use RESET SETTINGS to clear all MQTT settings.

Instructions:

- Connect the device to a Wi-Fi network and verify that a valid IP address is assigned.
- Enter the MQTT Broker address and Port number.
- Select the required Security mode (None/TLS).
- Enter the MQTT Topic, Device ID, Username, and Password (if authentication is enabled).
- Set the desired Publish Interval and select the time unit (Seconds or Minutes). The device will automatically publish sensor data to the MQTT cloud at the selected time interval.
- Click **SAVE CONFIGURATION** to store the settings.
- The device will automatically restart and reconnect using the new configuration.
- The default **MQTT** publish interval is **5 seconds**.
- The display refreshes sensor readings every **2 seconds** to provide near real-time monitoring.
- **IAQ**, sensors require approximately **5 minutes** of warm-up time after power-up before providing stable and accurate readings.
- Verify the cloud connection status to ensure sensor data is being published successfully.
- Click **RESET SETTINGS** to clear all saved MQTT configuration and restore the default cloud settings.

Contact us: -

**Augmatic Technologies Pvt. Ltd.,
Plot no 6, Shah Industrial Estate II,
Kotambi,
Vadodara – 391510.**

Email – Sales@wittelb.com

Support: support@wittelb.raiseaticket.com / +91 94038 92805